

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 4: Meiosis and Its Stages: Why Do Sperm and Egg Cells Have 23 Chromosomes?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	D	B	C	A	C	A

USE AFTER: PREDICT PHASE — Predict: chromosome-number problem (46 + 46 after fertilisation)

Q1. A human body cell has 46 chromosomes. Suppose a sperm cell and an egg cell each also had 46 chromosomes. How many chromosomes would the cell formed at fertilisation (the zygote) have?

- A. 46
- B. 92**
- C. 138
- D. 23

Correct answer: B. The zygote gets all the chromosomes from both gametes, so $46 + 46 = 92$. This would keep doubling every generation, which is why gametes must carry fewer chromosomes.

USE AFTER: OBSERVE PHASE — Observe: jigsaw sequencing and labelling of meiosis I pictures (Prophase I to Telophase I)

Q2. During meiosis I, which event happens in Anaphase I?

- A. Chromosomes copy their DNA again
- B. The cell membrane splits to make four cells at once
- C. Sister chromatids separate and move to opposite poles
- D. Homologous chromosomes separate and move to opposite poles**

Correct answer: D. In Anaphase I whole homologous chromosomes of each pair are pulled apart. Sister chromatids separate later, in Anaphase II.

USE AFTER: OBSERVE PHASE — Observe: Video 2 (meiosis II) and labelling of Prophase II to Telophase II

Q3. Which statement correctly describes Anaphase II of meiosis?

- A. Chromosomes line up as pairs at the middle of the cell
- B. Sister chromatids separate and move to opposite poles**
- C. Homologous chromosomes separate for the first time
- D. Homologous chromosomes pair up and cross over

Correct answer: B. Anaphase II is when the sister chromatids of each chromosome separate. Separating homologous chromosomes happens in Anaphase I, so mixing the two up is a common error.

USE AFTER: OBSERVE PHASE — Observe: recording chromosome numbers at the start, after meiosis I and after meiosis II

Q4. A human body cell with 46 chromosomes starts meiosis. What is found in each cell at the end of meiosis I (after Telophase I)?

- A. 46 chromosomes, each with one chromatid
- B. 46 chromosomes, each with two chromatids
- C. 23 chromosomes, each with two chromatids**
- D. 23 chromosomes, each with one chromatid

Correct answer: C. Homologous chromosomes separate in meiosis I, so the number halves from 46 to 23. Each chromosome still has two chromatids until they separate in meiosis II.

USE AFTER: EXPLAIN PHASE — Explain: discussion diamond ranking the four statements about why gametes have 23 chromosomes

Q5. Why do sperm and egg cells each have 23 chromosomes instead of 46?

- A. Meiosis halves the chromosome number, and fertilisation restores 46 in the zygote**
- B. Mitosis makes them, so they have the same number as body cells
- C. They have 23 because the zygote needs only 23 chromosomes to grow
- D. They have 46 chromosomes, and the extra 23 are thrown away after fertilisation

Correct answer: A. Meiosis reduces the chromosomes from 46 to 23 in each gamete. When a sperm (23) and an egg (23) join, the zygote has 46 again, so the number stays constant over generations.

USE AFTER: EXPLAIN PHASE — Explain: comparing meiosis and mitosis in a table (number of divisions, daughter cells, chromosome number)

Q6. Which row correctly compares meiosis in the reproductive organs with mitosis in the muscle cells of a Kenyan runner?

- A. Meiosis: one division, two identical cells with 46. Mitosis: two divisions, four different cells with 23
- B. Meiosis and mitosis both give two identical cells with 46 chromosomes
- C. Meiosis: two divisions, four genetically different cells with 23. Mitosis: one division, two identical cells with 46**
- D. Meiosis: two divisions, two identical cells with 46. Mitosis: one division, four cells with 23

Correct answer: C. Meiosis has two divisions and makes four genetically different cells with half the chromosomes (23), helped by crossing over. Mitosis has one division and makes two identical cells with 46 for growth and repair.

USE AFTER: MODEL BUILDING PHASE — Model: revising the class life-cycle model (zygote, repeated mitosis, growth to a teenager)

Q7. In a model of the human life cycle, a zygote with 46 chromosomes grows into a teenager. Which process produces the billions of new body cells for this growth, and how many chromosomes does each new body cell have?

- A. Mitosis, giving cells with 46 chromosomes**
- B. Fertilisation, giving cells with 92 chromosomes
- C. Meiosis, giving cells with 23 chromosomes
- D. Meiosis, giving cells with 46 chromosomes

Correct answer: A. The zygote divides repeatedly by mitosis, which makes identical cells with 46 chromosomes. Meiosis is used only to make gametes with 23 chromosomes.